

AINSHAMS UNIVERSITY

Faculty of Engineering

Computer and Artificial Intelligence Engineering

CSE245: Advanced Algorithms and Complexity



Parser – Team#2

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Terminals

Token classes	identifier					constant				stringLiteral			
Punctuation	:=		;		,	(		)		{		}	
Keywords	read	write	repeat	until	if	elseif	else	then	return	endl	end	main	

- **Datatype** → int | float | string
- **Mul_Operator** → * | /
- **Add_Operator** → + | -
- **Condition_Operator** → < | > | = | <>
- **Boolean_Operator** → && | " | "
- **Epsilon** → ε

Production Rules

1. High-Level Program Structure

- **Program** → **Function_Statements** **Main_Function**
- **Function_Statements** → **Function_Statement** **Function_Statements** | **Epsilon**
- **Function_Statement** → **Function_Declaration** **Function_Body**
- **Main_Function** → **Datatype** main () **Function_Body**
- **Function_Declaration** → **Datatype** identifier (**Parameters**)
- **Parameters** → **Parameter** **Parameter_List** | **Epsilon**
- **Parameter_List** → , **Parameter** **Parameter_List** | **Epsilon**
- **Parameter** → **Datatype** identifier
- **Function_Body** → { **Statements** **Return_Statement** }

2. Statements

- **Statements** → **Statement** **Statements** | **Epsilon**
- **Statement** → **Declaration_Statement** | **Assignment_Statement** | **Write_Statement** | **Read_Statement** | **If_Statement** | **Repeat_Statement**
- **Declaration_Statement** → **Datatype** **Decl_List** ;
- **Decl_List** → **Decl_Item** **Decl_List'**
- **Decl_List'** → , **Decl_Item** **Decl_List'** | **Epsilon**
- **Decl_Item** → identifier **Decl_Item'**
- **Decl_Item'** → := **Expression** | **Epsilon**
- **Assignment_Statement** → identifier := **Expression**
- **Write_Statement** → write **Write_Value** ;
- **Write_Value** → **Expression** | endl
- **Read_Statement** → read identifier ;
- **Return_Statement** → return **Expression** ;

- **Repeat_Statement** → repeat **Statements** until **Condition_Statement**

3. Control Flow

- **If_Statement** → if **Condition_Statement** then **Statements** **If_Tail**
- **If_Tail** → **Else_If_Statement** | **Else_Statement** | end
- **Else_If_Statement** → elseif **Condition_Statement** then **Statements** **If_Tail**
- **Else_Statement** → else **Statements** end

4. Expressions and Logic

- **Condition_Statement** → **Or_Condition**
- **Or_Condition** → **And_Condition** **Or_Condition'**
- **Or_Condition'** → “|” **And_Condition** **Or_Condition'** | Epsilon
- **And_Condition** → **Condition** **And_Condition'**
- **And_Condition'** → “&&” **Condition** **And_Condition'** | Epsilon
- **Condition** → **identifier** **Condition_Operator** **Term**
- **Expression** → **stringLiteral** | **Equation**
- **Equation** → **Multiplicative_Expr** **Equation'**
- **Equation'** → **Add_Operator** **Multiplicative_Expr** **Equation'** | Epsilon
- **Multiplicative_Expr** → **Primary_Expr** **Multiplicative_Expr'**
- **Multiplicative_Expr'** → **Mul_Operator** **Primary_Expr** **Multiplicative_Expr'** | Epsilon
- **Primary_Expr** → **Term** | (**Equation**)
- **Term** → **constant** | **identifier** **Term'**
- **Term'** → (**Arg_List**) | Epsilon
- **Arg_List** → **identifier** **Arg_Tail** | Epsilon
- **Arg_Tail** → , **identifier** **Arg_Tail** | Epsilon